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CSA1717 – ARTIFICIAL INTELLIGENCE

CO1 – LIST LAB PROGRAMS PYTHON

EXP 6: Vacuum Cleaner Problem

PROGRAM:

practical lab 6.py - C:/Users/yashm/OneDrive/Desktop/PYTHON AI/practical lab 6.py (3.14.5)

File Edit Format Run Options Window Help

```
rooms = {"A": "Dirty", "B": "Dirty"}
location = "A"

print("Initial state:", rooms)

for room in ["A", "B"]:
    location = room

    if rooms[location] == "Dirty":
        print("Cleaning room", location)
        rooms[location] = "Clean"

print("Final state:", rooms)
print("All rooms are clean.")
```

OUTPUT:

IDLE Shell 3.14.5

File Edit Shell Debug Options Window Help

```
Python 3.14.5 (tags/v3.14.5:5607950, May 10 2026, 10:43:50) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/yashm/OneDrive/Desktop/PYTHON AI/practical lab 6.py ====
Initial state: {'A': 'Dirty', 'B': 'Dirty'}
Cleaning room A
Cleaning room B
Final state: {'A': 'Clean', 'B': 'Clean'}
All rooms are clean.
>>>
```